

Raitu Vrudhi – An Android based Mobile Application for Agro-Marketing

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Abstract—The proposed work is a mobile application that helps farmers perform agro-marketing, leading to success and increasing their living standards. This work provides a solution for farmers in vain during the pandemic as they cannot sell the crop yields due to a lack of buyers and market agents who didn't buy the crop yields even at a quarter of the market price. Many farmers end up getting little pay for their hard work due to the ever-extending mediators in the value. The intuition of this novel real-time application is to improve communication between farmers and buyers easily with a constructive solution to revolutionize agricultural products for more returns on their yields. Considering the stated for this problem, an Android-based mobile agro-marketing application is developed that serves as a way for the farmers to sell their crop yields helps them to have more confidence in their sales, and will make farming more valuable by surpassing the mediator-dominated farming value chain. The mediators are replaced with a virtual system that provides direct interaction between farmers and markets or buyers with multiple options. A pre-requisite is just knowledge about the mobile and how to use the app to gain more insights into the farmer's side. This application addresses the communication gap faced by farmers during the pandemic despite having a high yield of crop harvest, but unable to sell it as they couldn't find any buyers and agents, to get market price and thereby increase their accumulation and growth. Thus, the mobile app offers a single platform that makes communication between the farmer and buyer straightforward and efficient aiming to help farmers grow economically.

Keywords—Agro-marketing, Mobile Applications, Android App, Direct Marketing

I. INTRODUCTION

Rural India is drastically moving towards digitalization and technology these days for a growing and prosperous India. The use of mobile technology in agriculture is increasing daily and, with these agro-marketing apps, farmers can get access to best farming practices, market prices, and more. Earlier, due to a lack of knowledge and the pandemic, farmers weren't able to bring more yield to their farms. The proposed Raitu Vrudhi is uploaded to the Google Play store and can be easily downloadable using a smartphone with the internet. By using this Raitu Vruthi app, farmers can now improve their farming practices, and have options for reviewing the best prices for their crops. The improvements to the work are to make the app available in popular regional languages of the country so that rural

farmers can easily use it without a challenge. The mobile application, titled Raitu Vrudhi devised in Telugu regional language, with a meaning stating farmer's development with prosperity. The Android App will serve as a way for farmers to sell their crop yields across the country. Profits will be maximized as customers interact directly with farmers and can negotiate based on user preferences. It helps farmers gain more confidence in their sales and will make farming more valuable than it is when lots of mediators kill the farming value chain.

The novel Raitu Vrudhi piloted mobile app helps farmers to sell fresh fruits and crop items across the country so that farmers can improve income gains from production gains for the best possible price, without having to take time away from farming. The application stimulates the farmers through the sharing of performance data and community feedback. Toward implementation, both farmers and buyers need to create an account and log in to the application. The farmer can post the details of his land and crop yield. These details include yield type, quantity, price, land type, etc. A buyer can search for the required item and contact the farmers directly. Once availed with the credentials for the site, the user can perform different operations like viewing the crops available to be procured, requesting or selling the yield, etc. Farmers can get in touch with buyers through the site, post their yield information and update their status of it. Buyers will be able to interact with the farmers easily without the help of the mediators. It keeps track of the history of the selling and buying processes, and suggestions can be made based on the history. A section in the app provides links for weather forecasts for farmers cultivating in dry and wetlands areas.

This Android app is a first-of-its-kind personal utility app made specifically for farmers provides a free, safe, and secure service, and provides crop and time-based profit/loss analysis. The app is easily accessible by Indian farmers by providing expert information on the Home page, Registration & Login page, Profile Page, Notification Page, Firebase Real-time Database, FAQs Page, Tenders Page, Product Page, Buyer Page, Seller page, Query Page, Products Track Page, Contact Page, All the pages hoarded in the app help farmers increase their farming productivity and yield. This agro app is a must-have for farmers who want to op to stay updated on the best prices for farming products. Farmers' apps have become the best solution to overcome farming challenges. With this app, farmers can now easily know

what's the best way to identify buyers, produce crops and sell the crop yields at best-selling rates.

Moreover, this Raitu Vrudhi farming app is used as a more convenient and useful medium to guide farmers in farming. Both buyers and sellers can access a variety of informative modules and get in touch with each other. This agro platform connects farmers to the world and provides them with all relevant agriculture-allied information.

Raitu Vruthi is a total farmer-friendly app that enables farmers to access all the information related to market prices of varieties of crop categories. The app has been developed to keep farmers abreast of crop prices and discourage them to go for distress sales. The app is nurtured with the basic principle of agriculture marketing, i.e., the selling of agriculture products by market price, and demand. depends on various components like the demand for the product at that time, availability of storage, increasing attention to ways of promoting direct linkages between farmers and buyers, etc. This helps the farmers to get a regulated price for the crop yield and realize the full potential of agricultural products and decide on a fair price. This initiative overcomes the role of mediators, who set back unequal coverage of farming prices, and inefficient financial management. All these measures were penned down to guard the income of the farmers and minimize the dominance of private traders in agricultural markets.

The objective of this agriculture app is to emphasize the agricultural sector needs to be with structured and functional markets, preferably in the vicinity of farmers, to drive growth, employment, remunerative price, and economic prosperity in rural areas of the country. Enabling mechanisms were also required to be put in place for the procurement of agricultural commodities directly from farmers' fields and to establish effective linkage between farm production, and the retail chain. The farmers can directly create Agricultural Marketing from where marketing impulses are transmitted to put all marketing interventions on track to safeguard the interest of both farmers and consumers. The app is focusing and taking concerted efforts to ensure remunerative prices to the farmers by creating marketing facilities.

The paper organization begins with an Introduction as Section 1 states the importance of the work, and Section 2 describes the background of this mobile application and the related work as a literature survey. Section 3, briefly explains the research methodology adopted in the work and Section 4 reports the implementation analogy findings of this work and KPIs used to validate the work. Finally, the article is concluded with possible future research extension to this work in Section 5.

II. LITERATURE SURVEY

The authors [1] in their published book make comprehensive developments happened in Agricultural marketing over the past 70 years. Categories in marketing functions are described in three ways as Exchange functions, Physical Functions, and Facilitating Functions. This work confines the usage of Exchange functions and its model. The authors of [2] adopt the e-trading model and drill up a framework that works in two phases. A case study on the national agriculture market was enabled and the impact was studied. The findings perceive ease of use of agricultural

marketing and improving the supply chain directly between buyer and sellers, which was adopted in this work.

The authors of [3] emphasize the importance of marketing infrastructure for agriculture and the farmer's role in dealing with decision-making and controlling the financial aspects of their harvest. In this work, after the seller procures the farming products from the buyer the financial aspect is taken as per this work [3]. The authors of [4] emphasize the roles of the federation of farmer-producer organizations, and how it addresses the challenges faced by supply chain management. Solutions to the problems in Post harvesting management and marketing the food commodities problems were discussed in the article. The paper of the authors [5] highlights the impact of the pandemic and lockdown on essential commodities, food crops, and vegetable marketing and the disruption of the supply chain markets. The findings state the importance of direct marketing. Hence with such findings, these works inculcate the idea into practice. The article of [6] brings the new ideology adopted by the government in India towards implementing agricultural reforms for solving problems plaguing agriculture such as low profitability. The model adopted in this article works with a solution in support to put into practice the reforms at the ground level.

The article [7] discusses the marketing approaches for selling and buying fruits and vegetables in Indian markets. Emphasis on the model of the Raitu Vrudhi app proposed in this work shifts the horticultural output on large scale by establishing communication between both ends and moving toward the stability of the economy. The study presented in [8] is in line with Indian agriculture, the route post-CoP 26, stated the effective mitigation measures and appropriate adaptation technologies in agriculture. The app presented in this work is in line with this initiative. The work presented by the authors in [9] embeds mobile application technology with voice-based interaction. Such technology will be included in this work, for future scope. The survey results in the paper presented by [10] is conducted in Andhra Pradesh, using a mobile application for fertilizer calculation, and cattle management, which directs this work to move forward to farmers for live applications as they were capable to use mobile technology. The authors developed and used a mobile app for product marketing and distribution, their results presented in [11] proved the highest level of results and with this motivation, this work replaces a model between farmers and buyers with direct marketing and emphasizes the value-based economy using the supply chain. The latest article released by [12] brings an Android app that provides agro-advisory mobile applications. This framework provides novel insights to all the farmers. It also supports the context that mobile technologies in agriculture are needed for developing nations.

The study undertaken by authors in [13] identifies geo-locations of diverse crops and their pattern of grow time to time using mobile technology and an app extensively used by farmers. Such pre-tested apps are more informative to do this extensive work using a mobile app to help farmers. New inculcations to this work will take to the usage of IoT devices in agriculture and control via the mobile app, reflecting farmers' productivity in increasing the irrigation system in crops as supported by the work mentioned by [14]. Another empirical study taken up by researchers in [15], investigates that female farmers can use efficiently mobile apps for

agriculture for their respective use. The work of [16], brings the sustainable factors for optimal utilization, development, and production of crops with a mobile app, especially used by farmers directly. The research direction was toward using AI in work to assist farmers in detecting the disease of the plant. Our work confines value chain management between the buyers and sellers but can adopt the work as future scope. The paper [17] addresses supply relationships among farmers and buyers with emerging mobile apps and suggested agricultural economies in rural areas. This work replicates a methodology approach to strengthen the direct supply chain to make farmers more profitable, and thereby improve the economic sustainability of farmers in India. The advent fact is several technologies with a mobile app, especially for agro-marketing and developments are synergistic in this fourth-generation revolution. Eight dimensions of agriculture technology were present in the article [18]. This work has a case study with research methodology for supply chain marketing as one of the dimensions mentioned in the work of [18]. The study in [19] explores the use of mobile phones for farm-related activities during the COVID-19 pandemic.

This work has a significant association with the development of strategies for marketing farm products marketing, exchanging information on food commodities, communicating among farmers and buyers, and other services for farming, which was one of the motivations for the present article. The article [20], states the research gaps on the impact of the pandemic on agriculture and the lives of farmers. It urges to invest in technological interventions to boost farm sustainability. Believing the IT infrastructure such as digital phones can help farmers to conduct business, the app was designed by addressing the value chain and such is practiced in this paper. The paper [21] is oriented toward the impact of the pandemic on the fishing sector in India and the measures taken to overcome it. The same set of mitigation measures is customized towards the food crops in this work, coordinating the farmers with the long-term needs of the supply networks.

III. METHODOLOGY

The proposed Raitu Vrudhi android mobile application aims to establish direct end-to-end communication between buyers and sellers by creating value chain management and encouraging agriculture marketing to be more profitable and sustainable. The methodology is structured in the form decentralized paradigm, with a multi-tier architecture with four interfaces accessible in the front end and data correspondence in the backend. The app is designed via a structural programming model using the top-down approach. The home page is linked up with the four independent modules as mentioned to seller, buyer, communication, and query interfaces. A security layer is embedded with two-level protection mechanism with password and OTP for user registration, authentication, verification, and validation, which acts as a prime security measure for the app. The first module is meant for farmers, where they can upload their crop yields and farming products. Once the upload is completed farmers can view the list of products uploaded. The farmers/sellers can fix the price of the crop yield directly and initiate the value chain business and start agriculture marketing with the tender option. Further, such products are tracked with my track option via the digital phone. The second module is the seller interface where the user or customers can directly interact with the farmers. They can

use the search product option for their respective query to find the agriculture product. They can crosscheck the price, and negotiate with farmers directly using the add tender option. Once, the best price is quoted from the seller they can buy the farming product and get the product and perform m online transaction to buy the product with the cart option. The core part is the end-to-end communication between the two parties both buyer and sellers. The correspondence to this module is taken with options like notification via text messages and direct interaction with the call option to complete the transaction by finalizing the tendering for procuring the products and have a win-win situation at both seller and buyer end with the best price in supply chain marketing.

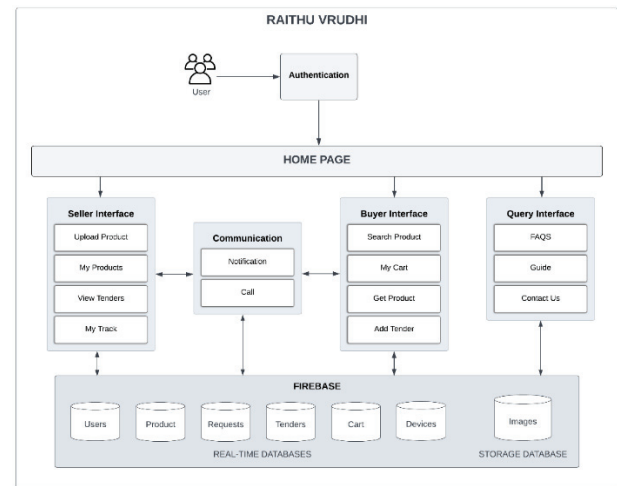


Fig. 1. Structural Architecture of Raitu Vrudhi Mobile Application

This result is the prime aim of the work since farmers get the best value for the product without misleading by the mediators. The fourth module is a query interface, which serves as an informative interface useful to novice users having a Guide option to use the mobile app, as the users are farmers. The other option is the FAQs answering the general queries raised operating the app and embedded with redirecting links to weather, market commodities prices, crop information, location, etc. Software debugging, testing, and

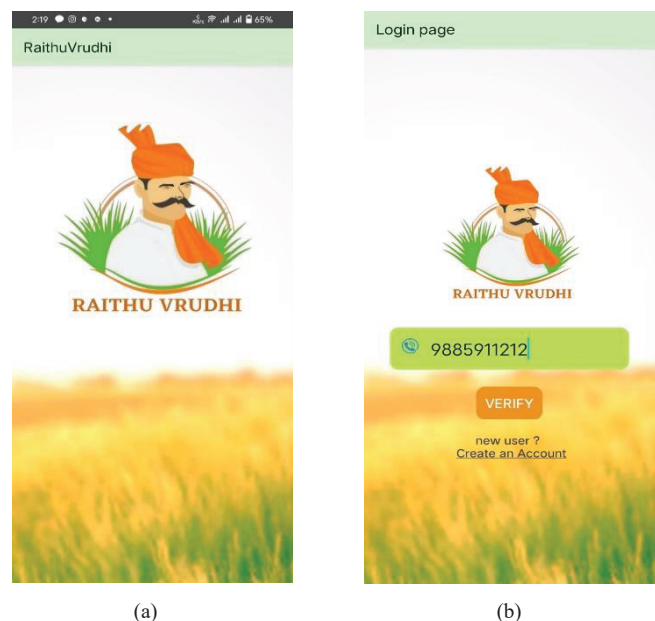


Fig. 2. Welcome screen of mobile app & Login screen of existing user

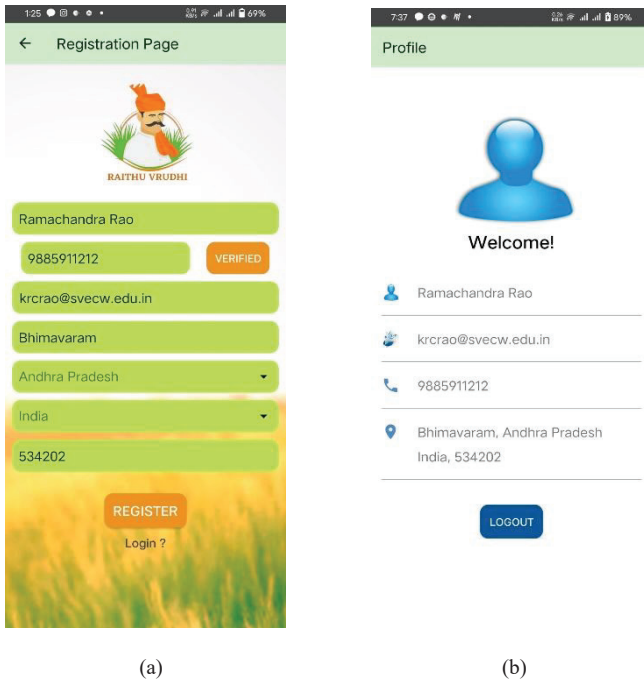


Fig. 3. Registration page for novice users & Profile page of existing user

IV. IMPLEMENTATION

This section elevates the usage of the fully functional real-time app Raitu Vrudhi implementation. The mobile app is implemented with Android Studio, supported with Java, XML, and Firebase as the backend database. The mobile app is hoisted with Welcome Screen as mentioned in Fig. 2 (a) and within 5 seconds, the screen is navigated to the Login page as mentioned in Fig. 2 (b). If the user is already existing in the App database, they can log in directly with his registered mobile number and password, otherwise, if it is a scenario with a novice user, a new account must be created and so it navigates to the Registration page as shown in Fig. 3 (a).

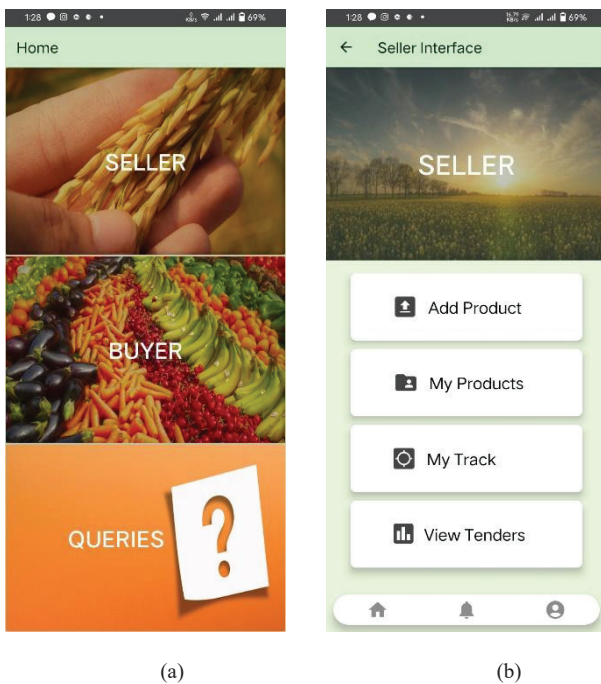


Fig. 4. Home page of the app with all modules & Seller interface screen

Authentication is provided to the novice user credentials with a verification code sent to the user's mobile in terms of One Time Password (OTP). Once the validation is completed the user registration is completed, and a profile page is created as shown in Fig. 3 (b) and will get complete access to all the modules mentioned in the mobile application.

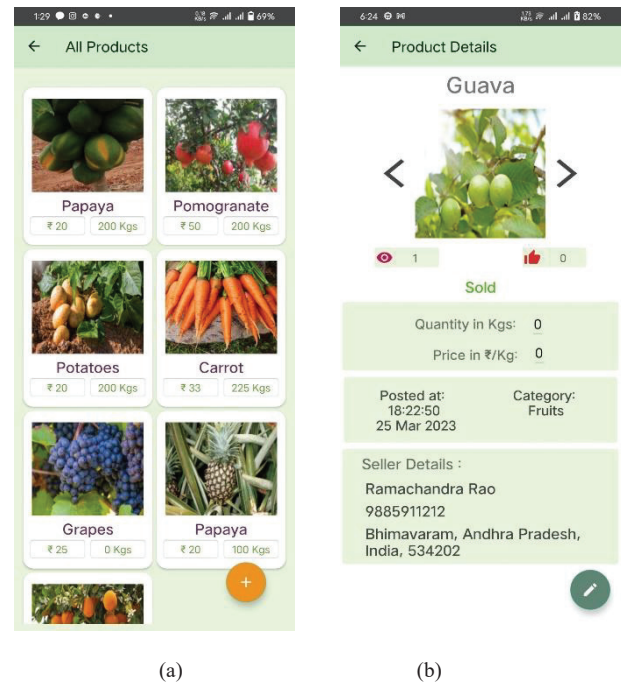


Fig. 5. Screen displaying agro products & Product details with status

The core part of the work is presented in Fig. 4 (a), named Home Page, by displaying the main modules of the Raitu Vrudhi mobile application. This page includes the Seller, Buyers, and Queries page interfaces, with a distinguishable look and feel, icons, and images. When moved to the seller Interface, it is operational with the sub-functional functionalities, Add Product, My Product, My Track, and View Tender options and is presented in Fig. 4 (b). The add product page represented in Fig. 5 (a), shows a form that has access from the seller or the farmer's end. Here the farmers can add their vegetables or fruit crop yields, the attributes include the price, weight in kilograms, and the product picture.

In extension, once the product is uploaded the status is marked as active, and immediate notification alerts are sent to the buyers. If the tender is approved and the procurement happens the status is changed to sold on the product details page, mentioned with time stamp, quantity, units, and seller information as shown in Fig. 5 (b). The work mainly emphasizes direct interaction between buyers and sellers with the end-to-end communication module. This communication system is implemented with two primitive attributes request and notifications as two-way handshaking and is shown in Fig. 6 (a). Once the farmer in the seller module uploads the crop details it is automatically intimated as a notification to all the registered sellers in the mobile app. The sellers may accept or decline the request and can call directly to the farmer to negotiate the price. The two communicating parties at both the end are the seller and buyer, from the home page of the app, it navigates to another important module called the Buyer interface, where the users or merchants can directly register and interact with the seller

and even have access to the posts and notification raised in the work.

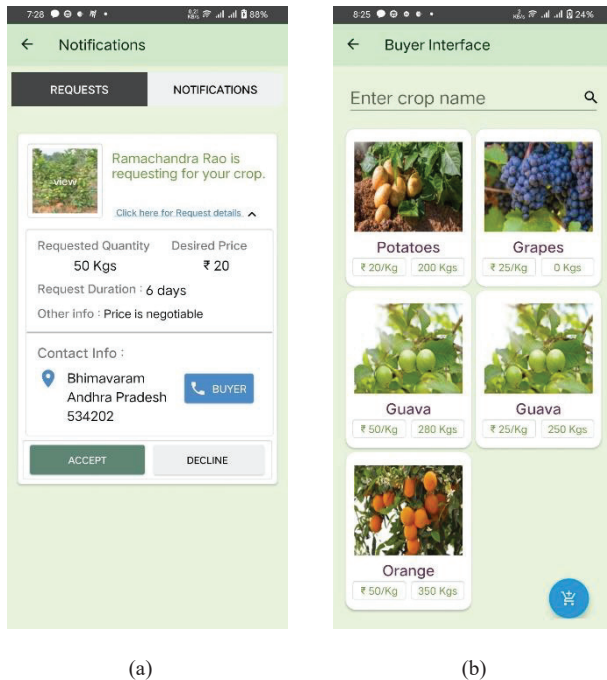


Fig. 6. Request and response notifications between farmers and buyers & Buyer interface with search option

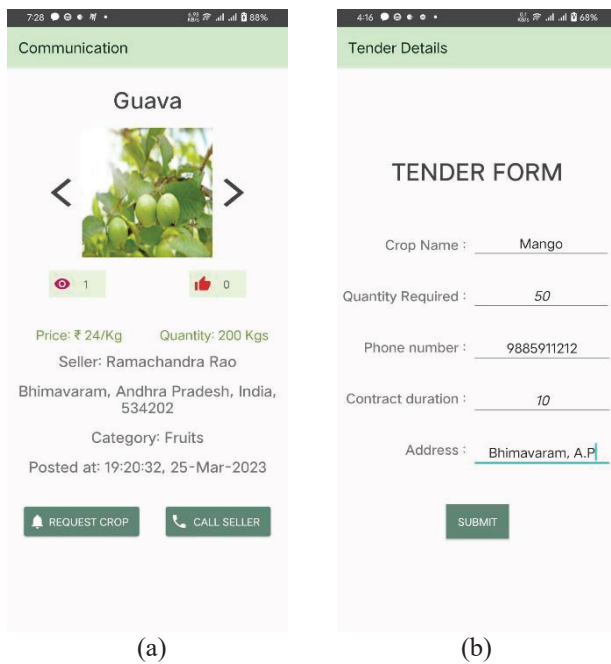


Fig. 7. Setting the buyer interface for communication & Partial procurement or price negation between parties using the tender page

The sub-functionalities associated with this mobile app interface were quick access for the buyer to search for a specific crop, call the farmer directly without the interference of the mediator, get the product details, tender for a crop, negotiate the best price, and finally purchase the crop yield directly from the farmer and adding it to my cart section. The search operation in the buyer interface is demonstrated in Fig. 6 (b). Once the buyer is interested to purchase the crop yield, the buyer can get into the crop page and get the crop

information via the communication notification with the seller, this usage is shown in Fig. 7 (a).

Along with this, the app provides an option to purchase partial purchase of fruits or vegetable crop products. In this case, the buyer can opt for a form to lay on a request for a specific crop, and if the crop is not yet available in the farmer notification or for an existing to make a partial procurement of the product the buyer will avail the tender request form. The tender is raised with the attributes stating the crop name, quantity of required crop in kilograms, contact number, and expected number of days the products are to be reached and the address location, the crop yield to be reached. The usage of this option is represented in Fig. 7 (b). After satisfactory communication is performed, the agriculture marketing reaches the final phase, where the products are placed in my cart window of the buyer, as shown in Fig. 8 (a), and is moved to the payment gateway for completing the online payment transaction.

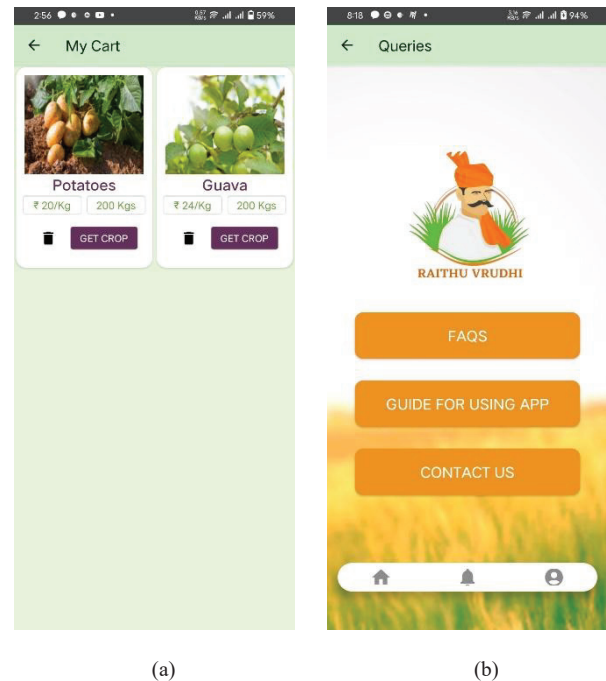


Fig. 8. Usage of my cart option facility in the buyer interface & Query page for buyers and sellers to effectively use the App

The generic page as shown in Fig. 8 (b) is the query page, which can be used by both the buyer and sellers for all purposes and with communication application developers, whether interfaces and demonstration page stating the steps and working functionalities of application interfaces and operational functionalities. The page shown in Fig. 9 (a) states the common queries and predefined answers that arise during the operation of the mobile and external API links to know weather forecasting, market commodities pricing, etc. Fig. 9 (b) hosts an exclusive demonstration page exclusively for farmers to make them learn how to use the Raitu Vruthi app with audio and video casting in regional languages etc. All such initiatives build confidence among farmers to be accustomed to using the app in a user-friendly manner and sustain better livelihood earning profits and directly being part of the development of the economy in the value chain markets.

A feedback mechanism helps the developers of mobile apps for future success and growth towards making a cost-effective model. It is possible by applying the review with the Key Parameter Indexes (KPIs) as evaluation metrics to access the performance of the mobile app and justify them from a technical standpoint. The app is recently uploaded to the play store and the download is quite reasonable because the technical understanding of front-end design is simple and customized from the perspective of farmers. The app is implemented on the Android operating system mobile architecture, and is relayed on gathering the Unique user ID, measures the count involved in screen views, and app speed. App load speed refers to the amount of time it takes before your app is ready for user interaction. Firebase is used to estimate the mobile app start time and is displaying as 221 milliseconds for this Raitu Vrudhi app. The app analytics used in this work, understand the user's operational device and operating system used. The aim is to focus on the quality of service to the user by catering to user needs. Users' sessions are recorded from the admin end by the unique user-id and the count has seen the number of users using the mobile OS. Screen resolutions also worked for this app, as Android phones may vary in size and shape. As farmers are the target audience, the query module has a guide and FAQ to support them, on how to use the app, and as part of maintenance, and troubleshooting happens in the app, a notification is sent to the developers as a crash report and appropriate action and intimation is taken. Other KPIs are related to analyzing the engagement of mobile users, to understand, how better they are using the app and retention. Google Analytics is performed over the app to understand the engagement of users with the mobile App. The session length is verified to estimate whether the sellers and buyers can open, conduct an activity, and close and the period of stay difference is calculated. Similarly, the session depth is estimated by whether the objective of the app is met, so that the farmers and buyers can communicate and do agriculture product direct marketing.

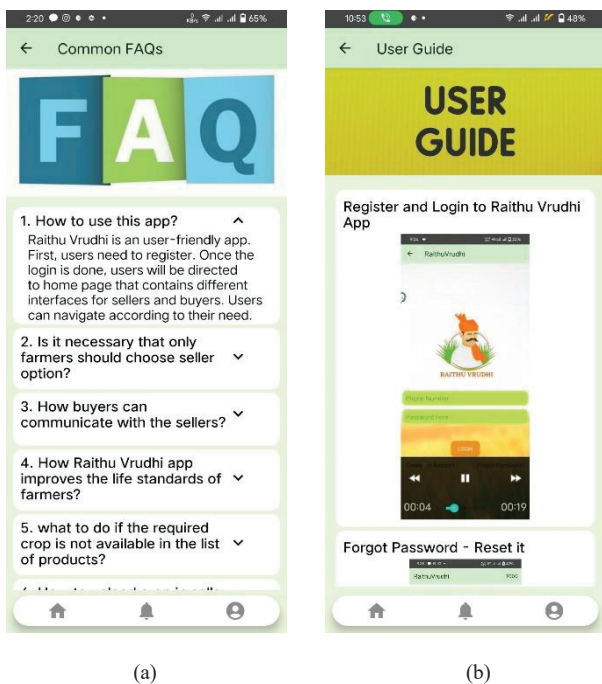


Fig. 9. FAQs and links to external API & User guide to farmers in the form of video and text demo

The farmers and buyers using the app can transit between the screen and are experiencing the right user-friendly design during the screens per visit. The facility to verify the unique users logged into the app monthly, weekly, and daily is also calculated for the mobile app. This initiative helps to understand the increase in popularity of this agro-marketing app in the play store. In due course of time, the user's stickiness ratio is estimated by verifying the usage of the user's log into the app regularly. An important observation is that due to the easy user-friendly interface design and app user guide video, the buyers and sellers who downloaded and installed from google play store retention are high and so the churn rate is very low. On a similar ground, the retention rate is high, since the monthly users who are using the app are aggressively using the app with a low churn rate. The growth rate is observed from the existing user's activity in using the app, the maximum time they are spending or using the app, it happens of their satisfaction in using the app, which means both farmers and buyers are making profits and smartly doing the agriculture business. Social Share to Raitu Vrudhi app is leveraged and the link is shared in the public domain as a service. To estimate the sign of engagement, the rating is considered as feedback and it is in a positive direction. The power factor plays are the faster app load speed and response time.

V. CONCLUSION AND FUTURE SCOPE OF WORK

The work addresses the issues raised by the Covid pandemic, where farmers are unable to sell their agricultural products for the best price and incurred huge losses in terms of money and time. The Raitu Vrudhi app is the first of its kind customized to Indian farmers, and played a crucial role in improving the livelihood of a farmer, by establishing a firm real-time network of communication between sellers and buyers through the app right at their fingertips. Farmers can gain profits and have the convenience to sell their agricultural products without any involvement of middlemen.

In a nutshell, farmers are assumed as sellers and can sell food commodities across the country with the buyers via this digital platform. The mobile app provides an opportunity to conduct online marketing of agricultural products. The realistic implementation is taken up mainly with four interfaces buyer, seller, communication, and queries. The welcome screen provides security and authentication for existing users and accepts registrations for new users with a verification and validation process. The farmer uses the seller's module and posts the crop information and can quote price tenders with sellers directly. The notification module acts as an interface between the buyers and sellers and displays notifications of agriculture products uploaded by farmers on the other end. The buyers can view notifications, search marketing products and negotiate pricing with the farmers directly by dialing or messaging without the intervention of a third party. Once the deal is finalized the crop yields are moved into my cart option as per the requirement of the buyer and the payment gateway option is used for online money transfer to dispatch the product. The autonomous module is the query module that is helpful to both the buyers and sellers, which educates about the usage of mobile applications, videos are hosted as guides to the farmers so that they can watch and understand the usability of sub-functionality options in the app. External API links are given for weather forecasting, commodity Market rates,

etc. This mobile application created a facility for farmers to avail an opportunity to do direct marketing and sell their agricultural products and earn profits without the intercession of mediators.

Ultimately this platform minimized the role of middlemen, maintains transparency to the operations in food supply chains, and make the country's economy stronger in the agriculture sector. The extension and future scope of this work were to upgrade the mobile app to both Android and iOS versions and with new inclusion features like analytics, regional language correspondence, agricultural advisory, GPS, voice search, external links to crop insurance, online trading of seeds, machinery, fertilizers, etc.

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